

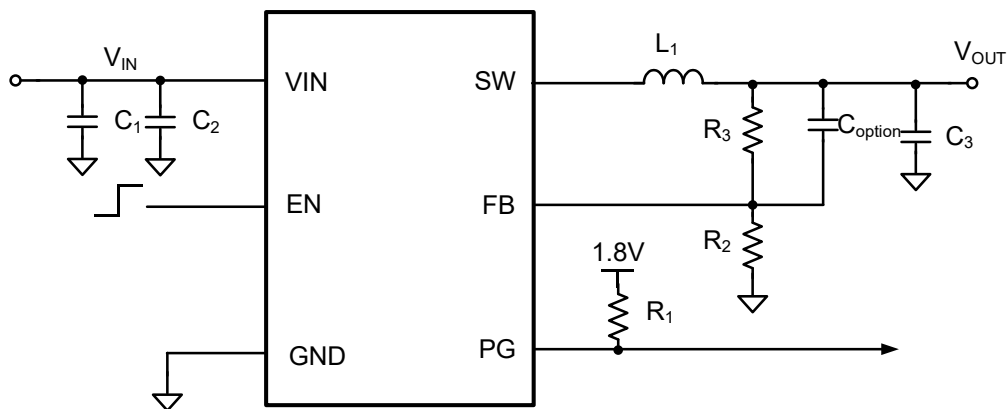
ET83102H – High Efficiency, 6V/1.5A Buck Converter

General Description

The ET83102H device family is a high-efficiency, synchronous step-down converter supports 1.5A output current and withstands 6V input voltage. The device operates with constant off time combined with a peak current control scheme. The control circuit calculates the off time for the low-side MOSFET, making the switching frequency relatively constant regardless of the variation of the input voltage, output voltage, and load current. The control scheme provides a fast transient response and eases loop stabilization. The 1.5MHz switching frequency allows the device to use small external components with minimum 10 μ F output capacitor.

The ET83102H supports 100% duty mode when the required output voltage is higher than the input voltage and smooth switch to step-down operation when the input voltage is higher than the target output.

The ET83102H is available in multiple packages: SOT563, SOT23-5 and SOT23-6.



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Features

- Input voltage range: 2.2V – 6V, down to 1.9V after startup
- Reference voltage accuracy: 0.6V $\pm$ 1.5% at room temperature
- 40 μ A typical quiescent current
- High efficiency operation
 - 92% efficiency at 1.2A load from 5V to 3.3V conversion
 - 86% efficiency at 0.2A load from 3.3V to 1.2V conversion
- Automatic PFM mode to extend battery life or Forced PWM version to minimize output ripple
- Typical 1.5-MHz switching frequency and 10 μ F output capacitor
- Constant off time control scheme to achieve fast load transient response
- 100% duty mode
- 1.2V IO
- PG output
- Integrated UVLO, OCP, SCP, and thermal protections
- 0.35ms typical soft-start time
- Active output discharge
- Package:

Part No.	Package	MSL
ET83102H	SOT563	Level 1
ET83102HT	SOT235	Level 3
ET83102HE	SOT236	Level 3

Applications

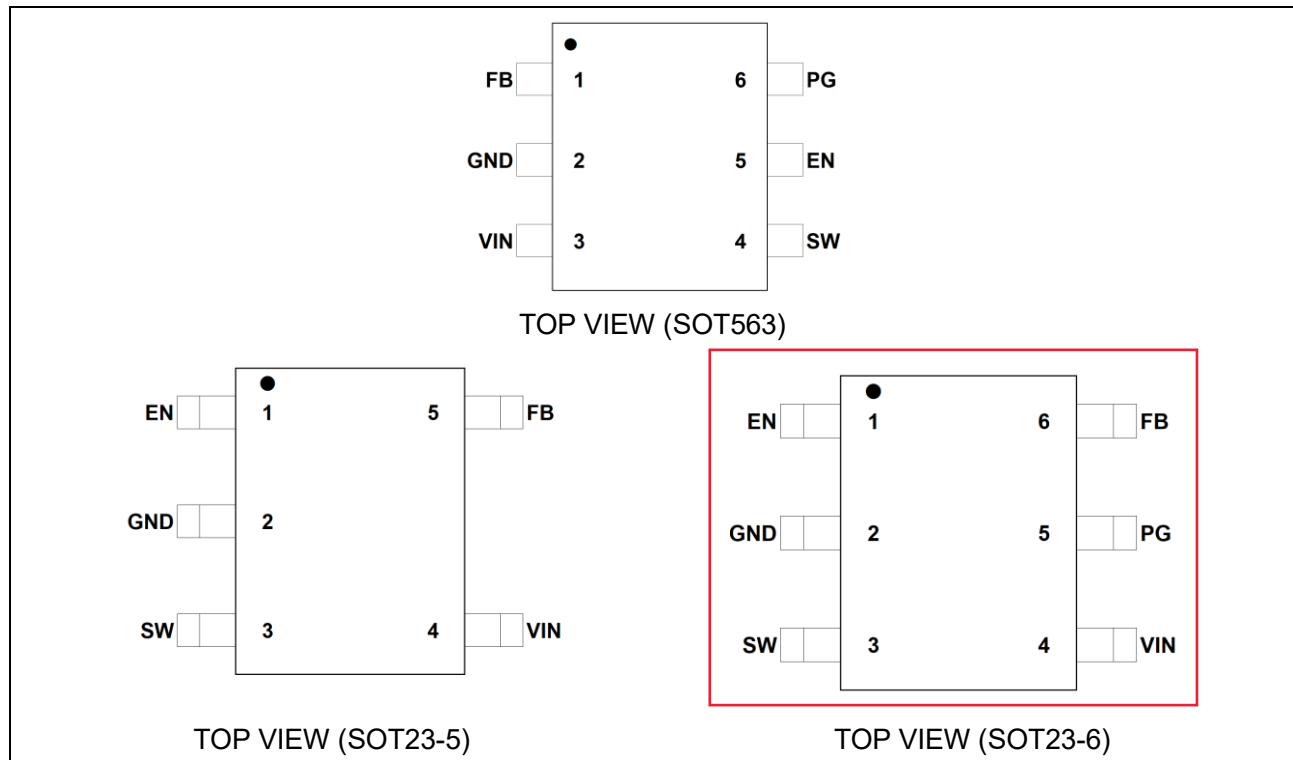
- IPC
- TV/Box
- USB or single-cell Li-ion battery applications

Device Information

Part No.	Package	Mode	Power Good
ET83102H	SOT563	Auto PFM	YES
ET83102HT	SOT23-5	Auto PFM	NO
ET83102HE	SOT23-6	Auto PFM	YES
ET83102HC	SOT563	FPWM	YES
ET83102HTC	SOT23-5	FPWM	NO
ET83102HEC	SOT23-6	FPWM	YES

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Pin Configuration

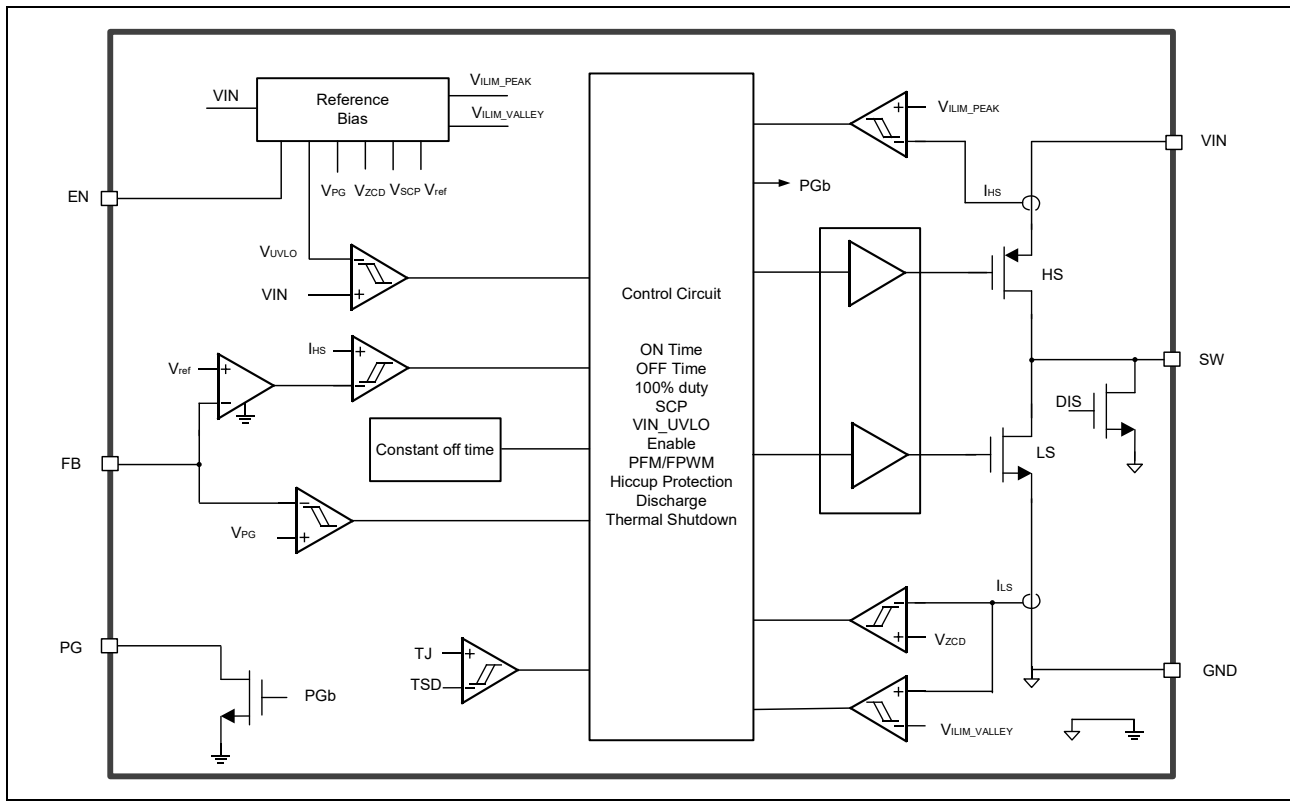


Pin Function

Pin No.			Pin Name	Pin Function
SOT563	SOT235	SOT236		
1	5	6	FB	Output voltage feedback. Connect resistor divider at this pin
2	2	2	GND	Ground
3	4	4	VIN	Power input. A 10 μ F bulk capacitor should be placed between this pin and GND pin.
4	3	3	SW	Switching Node
5	1	1	EN	Enable
6	/	5	PG	This pin is the power good indication. This pin is open-drain output.

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Block Diagram



Functional Description

Overview

The ET83102H is a high-efficiency synchronous step-down converter supports up to 1.5A output current and withstands 6V input voltage. The ET83102H operates with constant off time combined with a peak current control scheme to achieve fast transient response. The control circuit calculates the off time for the low-side MOSFET, making the switching frequency relatively constant regardless of the variation of the input voltage, output voltage, and load current. The ET83102H seamless transfer between the buck mode and 100% duty mode by sensing the input voltage and output voltage.

Constant OFF-Time Control Scheme

The ET83102H is synchronous buck converter with the constant-off-time (COT) control mechanism to provide fast transient response and eases loop stabilization. At the beginning of each cycle, the high-side MOSFET (HS-FET) is turned on. The on period is determined by both the error amplifier output voltage and the current through the HS-FET. When the current reaches the error amplifier output voltage, the on time is terminated and the LS-FET is turned ON, which is the start of the off-Time. The off time is calculated by the control circuit which is proportional to the voltage difference between the input voltage and output voltage. The HS-FET turned on again when the off-time timer is over. By repeating operations this way, the converter regulates the output voltage.

Internal compensation is applied for constant off time control for stable operation even when ceramic capacitors are used as output capacitors.

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The ET83102H operates at typical 1.5MHz switching frequency and supports 10 μ F output capacitor. Higher switching frequency operation enables better transient performance and reduced the output ripple at the cost of higher switching loss.

Auto-PFM

The ET83102H is designed to maintain high efficiency at light load by adopting automatic pulse-frequency modulation (Auto PFM). When the output current decreases, the current through the inductor decreases to zero during the off time. When the inductor current reaches zero, detected by the zero-current detection (ZCD) comparator, the low-side MOSFET is turned off, together with the high-side MOSFET. Both MOSFETs remain off until the amplifier output voltage rises above PFM threshold. As the load current decreases, the duration for both MOSFETs to remain off increases, leading to a lower switching frequency and higher power efficiency.

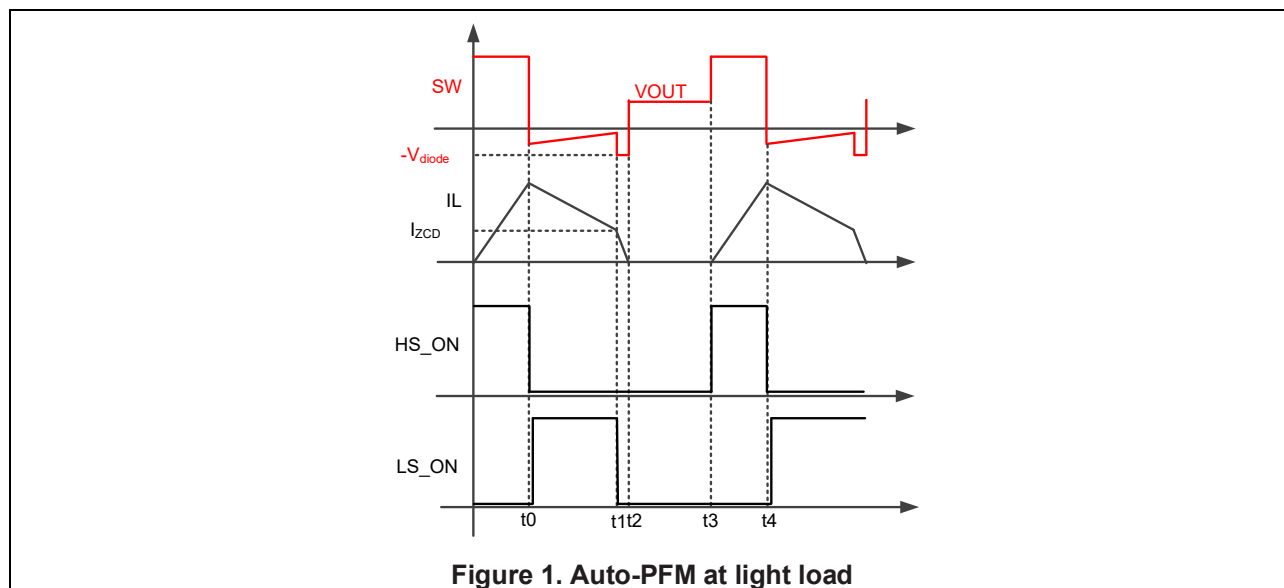


Figure 1. Auto-PFM at light load

The ET83102H exits the PFM mode automatically and enters CCM when the load increases to the level when the minimum inductor current increases to higher than zero.

FPWM (ET83102HxC)

In FPWM mode, the ET83102H keeps the switching frequency constant at light load condition. When the load current decreases, the high-side switch is not turned off even if the current of the high-side switch goes negative to keep the frequency constant. The output ripple is greatly reduced at light load compare with the PFM mode.

The ZCD threshold changes to positive with a typical -1.2A current limit which is named negative current limit (NOCL). When the inductor current hits the NOCL, the low side FET is turned off immediately and the high-side MOSFET is forced turning on after the dead time. In the next switching cycle, the inductor current ramps up again and then ramp down the same as the beginning.

For FPWM version, please choose the correct inductor to ensure the peak-to-peak current under no load condition is higher than 300mA.

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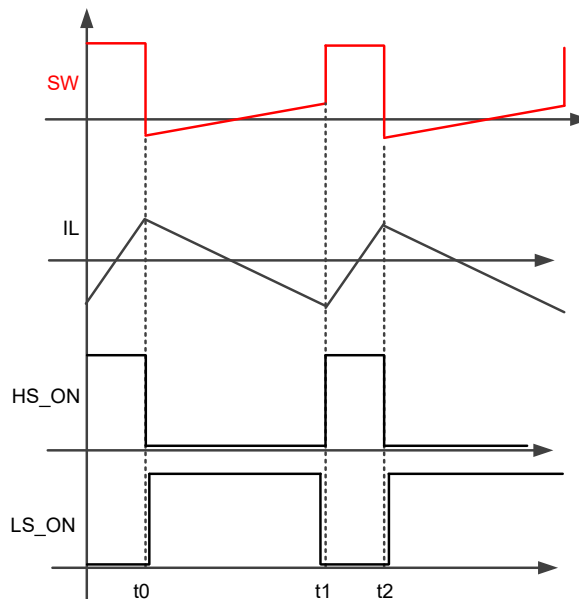


Figure 2. FPWM at no load

100% duty mode

The ET83102H supports 100% duty mode when the input voltage is close or even lower than the target output voltage. This is achieved automatically inside ET83102H. The HS-FET is kept always ON and LS-FET is always OFF. The current limit function and short circuit protections are still active in 100% mode.

Soft-start

When the VIN is above the UVLO threshold and EN voltage rises above the EN pin logic high threshold, after an EN blanking time t_b (1 μ s typically) and delay time T_d (see the value in EC table), then the ET83102H enables all the internal circuits, delays for a certain delay time, and begins the soft-start. The internal reference voltage ramps up to V_{ref} in t_{ss} time.

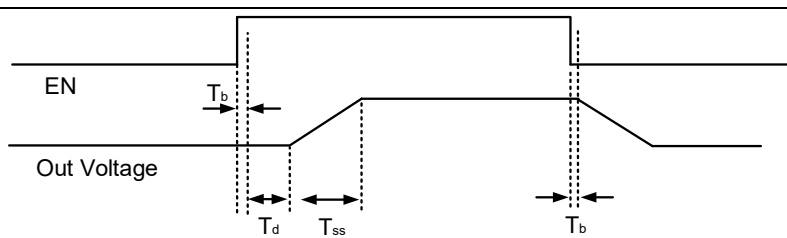


Figure 3. Soft-start and Shutdown

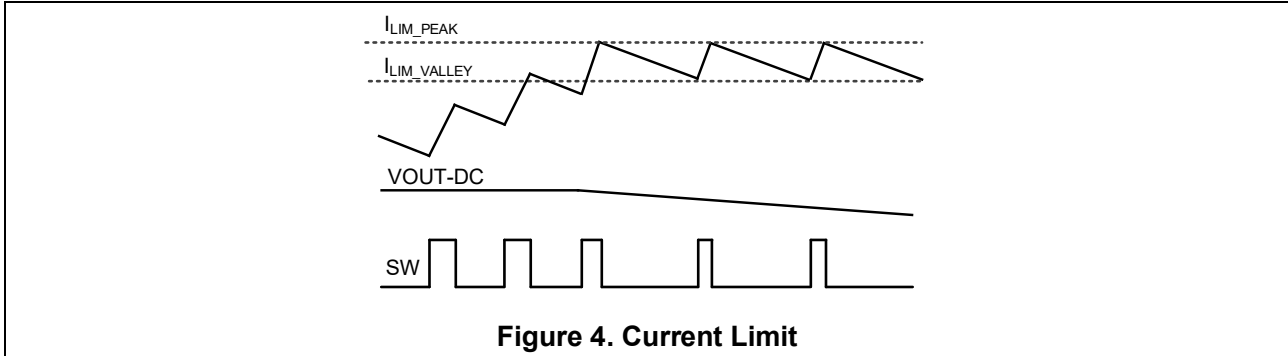
Over Current Protection and Short Circuit Protection (SCP)

The ET83102H protects an over current situation by limiting both the inductor both peak and valley current. The currents through HS-FET and LS-FET are monitored all the time when the ET83102H is enabled. The HS-FET is turned off immediately when the peak current reaches the peak current limit and the HS-FET cannot be turned on if the LS-FET valley current is higher than the low-side valley current limit.

The SCP is realized by monitoring the inductor current whether is limited. Once the output load draws more

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current than the current limit, the output voltage drops. When the peak current limit is triggered for 32 switching cycles, the ET83102H shuts down, the LS-FET is kept on until the ZCD threshold is reached. A new soft-start begins after a typical 130 μ s wait time.



Output Active Discharge Function

The ET83102H sinks current from the SW pin when the device is shut down by EN. An internal discharge FET is turned on and the energy from the output capacitor is discharged to the ground. The typical resistance of the FET is 15 Ω . The discharge function is disabled at below conditions:

1. During soft start
2. VIN UVLO threshold
3. Thermal shutdown

Power Good Indication

The Power Good (PG) pin is an open-drain output. Once the voltage of the FB pin rises above 93% of the target output voltage, the PG detection is triggered and PG pin floats after a 20- μ s deglitch time. A pullup resistor of 100 k Ω is recommended to pull the voltage up to VIN or VOUT. The PGD pin is pulled low when:

- FB pin voltage is lower than 90% of the target output voltage
- In a VIN UVLO, or thermal shutdown event
- During the soft-start period
- EN is pulled low

Thermal Protection

The ET83102H will shut down when the internal control circuit temperature is higher than 165°C and will restart after the temperature drops below 135°C as well.

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Absolute Maximum Ratings

Symbol	Parameters (Items)	Value	Unit
V _{IN}	Supply voltage	-0.3 to 7	V
V _{SW1(DC)}	Switching node voltage (DC)	-0.3 to (V _{IN} +0.3V)	V
V _{SW1(10ns)}	Switching node voltage (10ns transient)	-3 to 8.5	V
	All other pins	-0.3 to 6	V
P _D	Continues power dissipation	0.6	W
T _J	Junction temperature	-40 to 150	°C
T _{Solder}	Maximum Soldering Temperature (at leads-10s)	260	°C
T _{STG}	Storage temperature	-55 to 150	°C

ESD Ratings

Symbol	Parameters (Items)	Value	Unit
HBM	Human body model, JS-001	±2000	V
CDM	Charged-device model, JESD22-C101	±1000	V

Thermal Characteristics

Symbol	Package	Parameters	Value	Unit
R _{θJA}	SOT563	Junction-to-Ambient Thermal Resistance, JESD51-7	280	°C/W
	SOT563	Measured on EVM, 2-layer, 2oz copper	110	°C/W
	SOT23-6	Junction-to-Ambient Thermal Resistance, JESD51-7	200	°C/W
	SOT23-6	Measured on EVM, 2-layer, 2oz copper	80	°C/W
R _{θJB}	SOT563	Junction-to-Bottom Thermal Resistance, JESD51-7	80	°C/W
	SOT23-6		60	°C/W

Recommended Operating Conditions

Symbol	Parameters	Min	Max	Unit
V _{IN}	Input voltage	2.2	6.0	V
V _{OUT}	Output voltage	0.6	V _{IN}	V
T _A	Ambient temperature range	-40	85	°C
C _{IN}	Input decoupling capacitor, effective	4.7		μF
C _{OUT}	Output capacitor, effective	4.7	100	μF

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Electrical Characteristics

The specifications are measured under conditions $V_{IN} = 5V$, $T_J = 25^\circ C$, unless otherwise specified

Symbol	Parameters	Conditions	Min	Typ	Max	Unit
Input Section						
V_{ULVO}	Under voltage lockout threshold	Rising threshold		2.2	2.4	V
		Falling threshold		1.9		V
V_{OVP}	Over voltage protection threshold	Rising threshold	5.7	6.0	6.3	V
		Falling threshold		5.7		V
I_q	Input quiescent current	PFM, EN=VIN, FB=0.65V		40	65	μA
		Forced PWM, EN=VIN		0.5	2	mA
I_{IN_SD}	Input shutdown current	EN=0		0.1	5	μA
Power Stage						
I_{LIM_PEAK}	High-side peak current limit	$T_J=25^\circ C$	1.8	2.5		A
I_{LIM_VAL}	Low-side valley current limit	$T_J=25^\circ C$	1.4	2.0		A
R_{ON_HS-FET}	High-side FET on-resistance	$V_{IN}=5V$		210		m Ω
R_{ON_LS-FET}	Low-side FET on-resistance	$V_{IN}=5V$		150		m Ω
Control Circuits						
V_{ref}	Reference Voltage	$25^\circ C$	0.591	0.6	0.609	V
		$T_J=-40^\circ C \sim -125^\circ C$	0.582	0.6	0.618	V
I_{leak_FB}	FB leakage current				0.1	μA
F_{sw}	Switching frequency	1.5MHz		1.5		MHz
t_d ^[Note2]	EN delay time	From EN turn high to 1 st switch		30		μs
t_{ss} ^[Note2]	Soft start time	5% to 90% of output voltage		0.35	0.6	ms
EN Logic						
V_H	Logic high	Rising threshold	1.2			V
V_L	Logic low	Falling threshold			0.4	V
R_{EN}	EN pull down resistor			1		M Ω
$R_{DIS-FET}$	Discharge FET resistance	EN=0, 10mA, From SW to GND		12	20	Ω
Power Good						
V_{UVLO_rise}	PGD UV threshold	FB Rising threshold	88	93	98	%
V_{UVLO_fall}		FB Falling threshold	86	91	96	%
R_{PGD}	PG pull down resistance	1mA, From SW to GND		500	1000	Ω

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Electrical Characteristics (Continued)

Symbol	Parameters	Conditions	Min	Typ	Max	Unit
Short Circuit Protection						
t_{SCP} [Note2]	SCP deglitch time			32		cycles
t_{HICCUP} [Note2]	Hiccup wait time			140		μs
Thermal Shutdown						
T_{jSD}	Thermal shutdown threshold	Rising threshold	155	165	175	$^{\circ}C$
		Falling threshold		135		$^{\circ}C$

Note2: Not production tested. Guaranteed by design.

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Typical Characteristics

$V_{IN} = 5.0V$, $V_{OUT} = 1.2V$, 1.5MHz, HTTH25201T-1R5M (1.5 μ H, DCR=35m Ω).

Figure 5. V_{ref} vs T_J	Figure 6. Quiescent Current vs T_J , EN=VIN
Figure 7. Shutdown Current vs T_J	Figure 8. Enable/Disable threshold vs T_J

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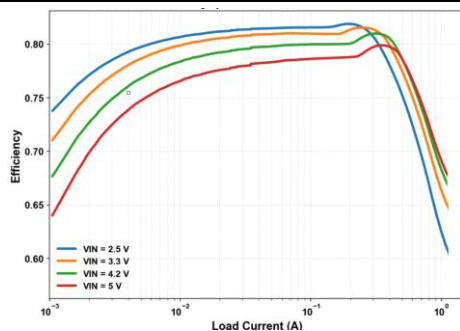


Figure 9. Efficiency vs Load Current,
 $V_{OUT} = 0.6V$, PFM

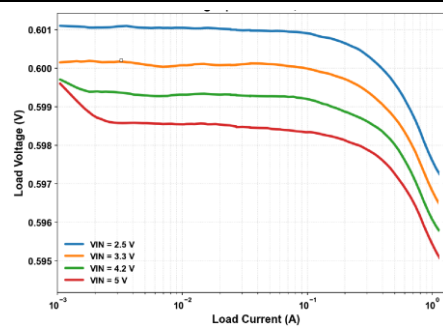


Figure 10. Load Regulation,
 $V_{OUT} = 0.6V$, PFM

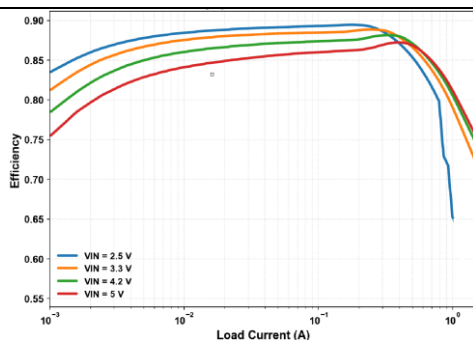


Figure 11. Efficiency vs Load Current,
 $V_{OUT} = 1.2V$, PFM

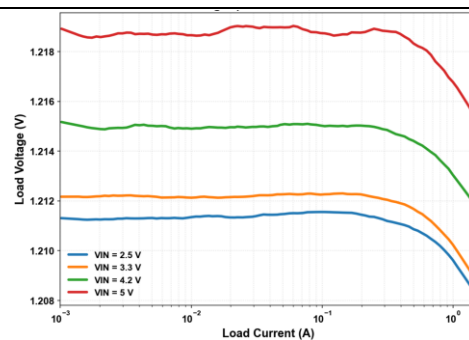


Figure 12. Load Regulation,
 $V_{OUT} = 1.2V$, PFM

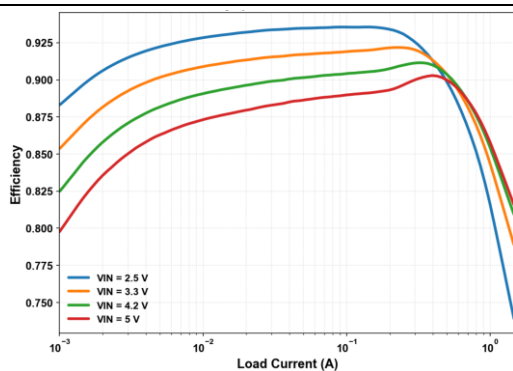


Figure 13. Efficiency vs Load Current,
 $V_{OUT} = 1.8V$, PFM

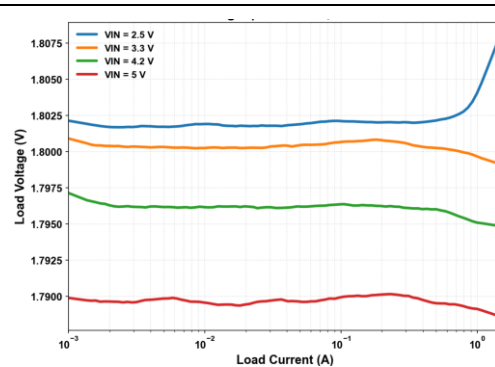


Figure 14. Load Regulation,
 $V_{OUT} = 1.8V$, PFM

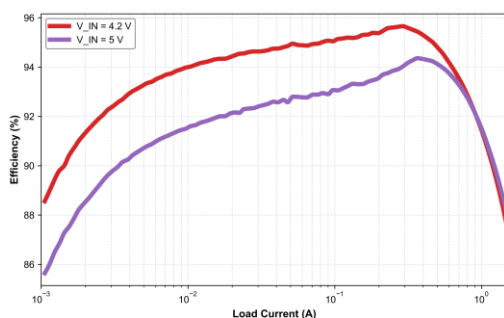


Figure 15. Efficiency vs Load Current,
 $V_{OUT} = 3.3V$, PFM

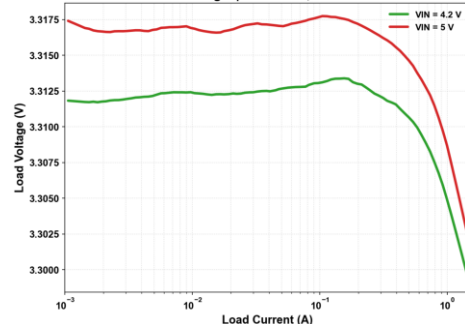


Figure 16. Load Regulation,
 $V_{OUT} = 3.3V$, PFM

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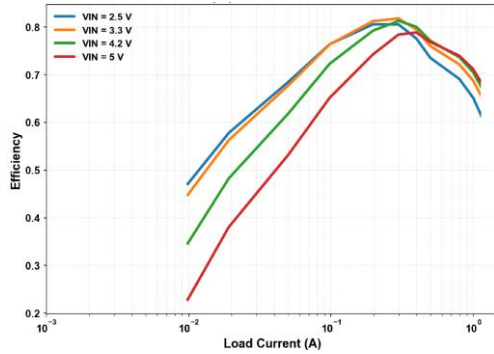


Figure 17. Efficiency vs Load Current,
 $V_{OUT} = 0.6V$, FPWM

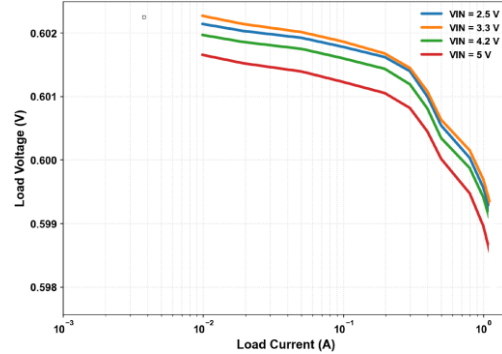


Figure 18. Load Regulation,
 $V_{OUT} = 0.6V$, FPWM

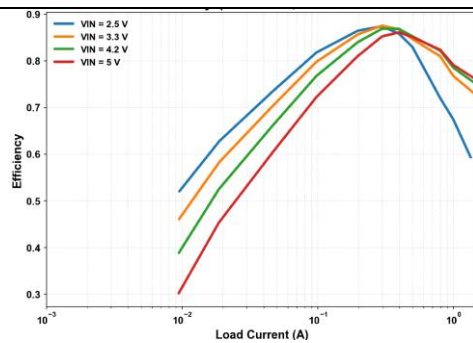


Figure 19. Efficiency vs Load Current,
 $V_{OUT} = 1.2V$, FPWM

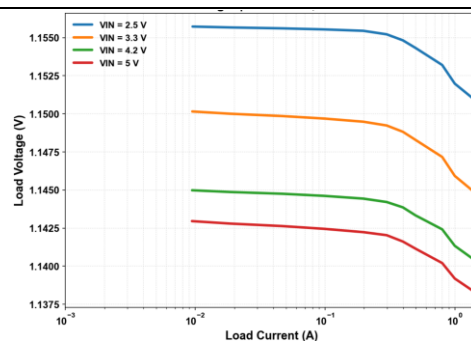


Figure 20. Load Regulation,
 $V_{OUT} = 1.2V$, FPWM

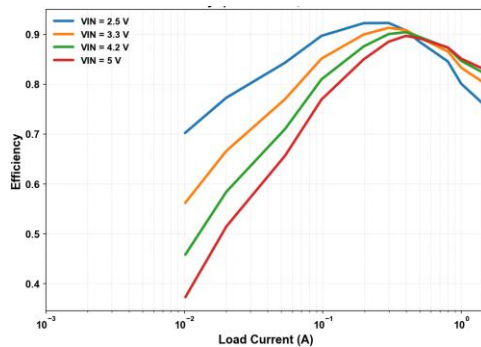


Figure 21. Efficiency vs Load Current,
 $V_{OUT} = 1.8V$, FPWM

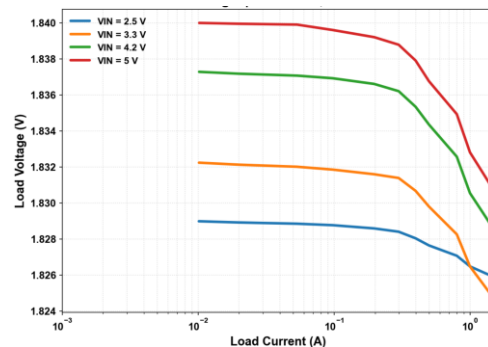


Figure 22. Load Regulation,
 $V_{OUT} = 1.8V$, FPWM

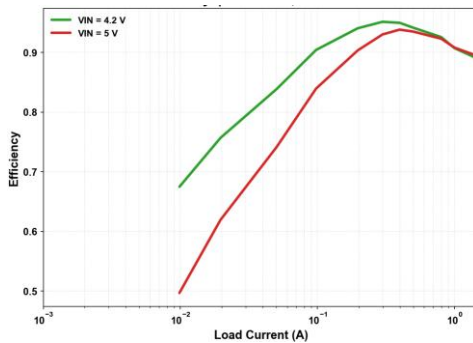


Figure 23. Efficiency vs Load Current,
 $V_{OUT} = 3.3V$, FPWM

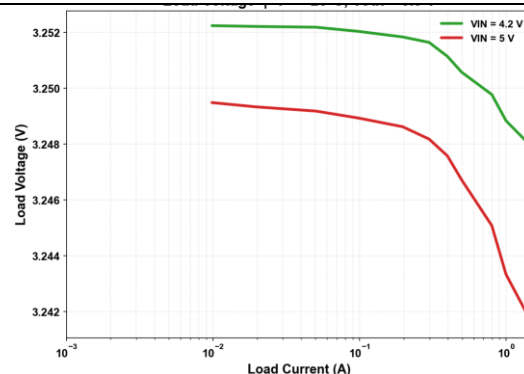


Figure 24. Load Regulation,
 $V_{OUT} = 3.3V$, FPWM

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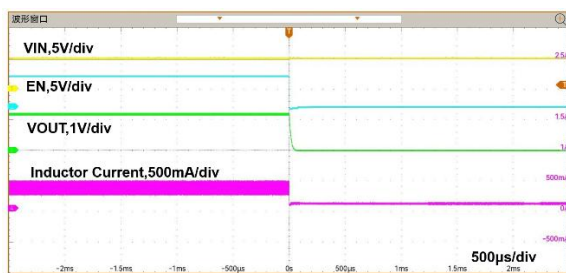


Figure 25. Output Ripple, 1mA load, PFM

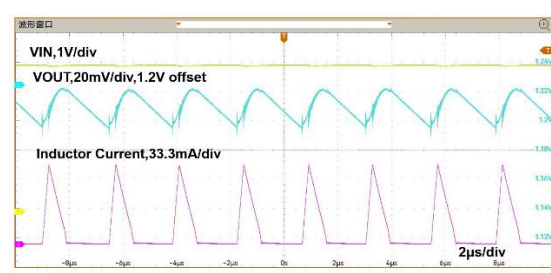


Figure 26. Output Ripple, 100mA Load, PFM

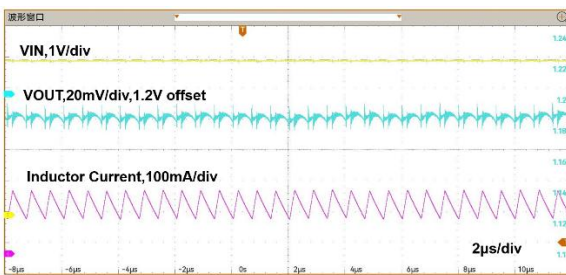


Figure 27. Output Ripple, 500mA Load

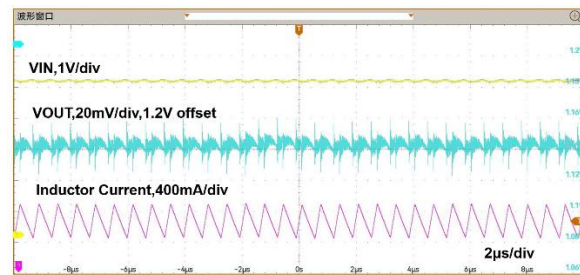


Figure 28. Output Ripple, 1.5A Load

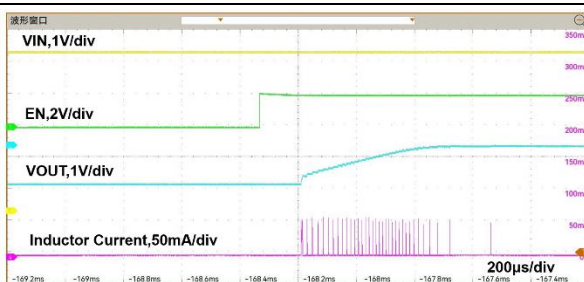


Figure 29. Startup by EN, No load

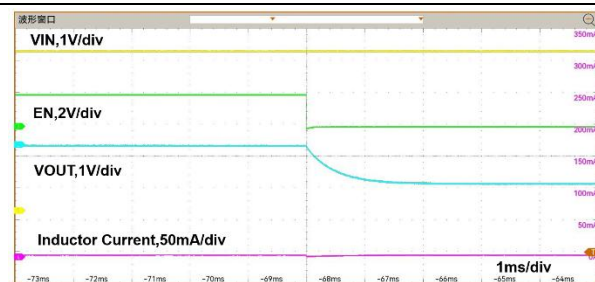


Figure 30. Shutdown by EN, No load

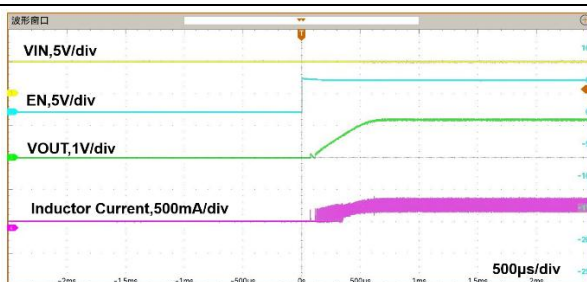


Figure 31. Startup by EN, 2.2Ω Load

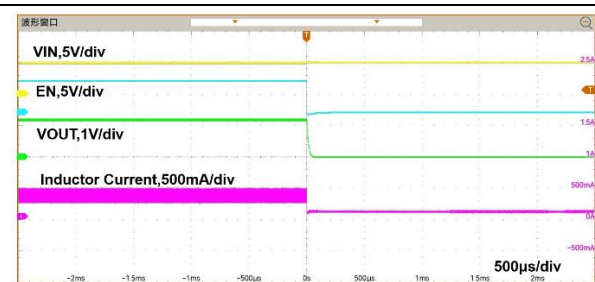


Figure 32. Shutdown by EN, 2.2Ω Load



Figure 33. Load Transient, $I_{OUT}=1\text{mA}$ to 0.5A, 1A/µs



Figure 34. Load Transient, $I_{OUT}=0.1\text{A}$ to 1.5A, 1A/µs

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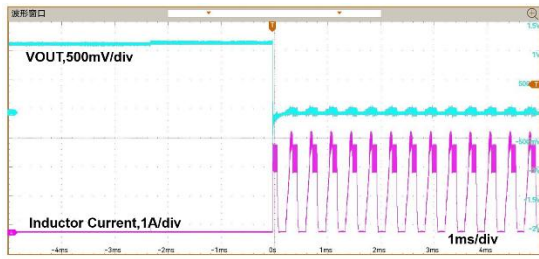


Figure 35. Short Circuit Protection

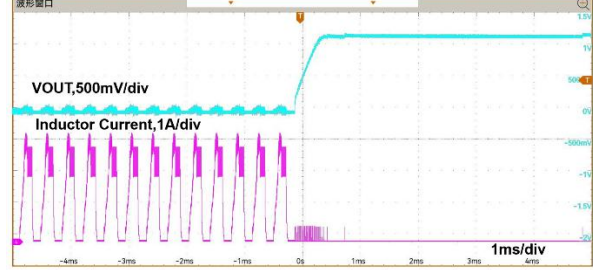
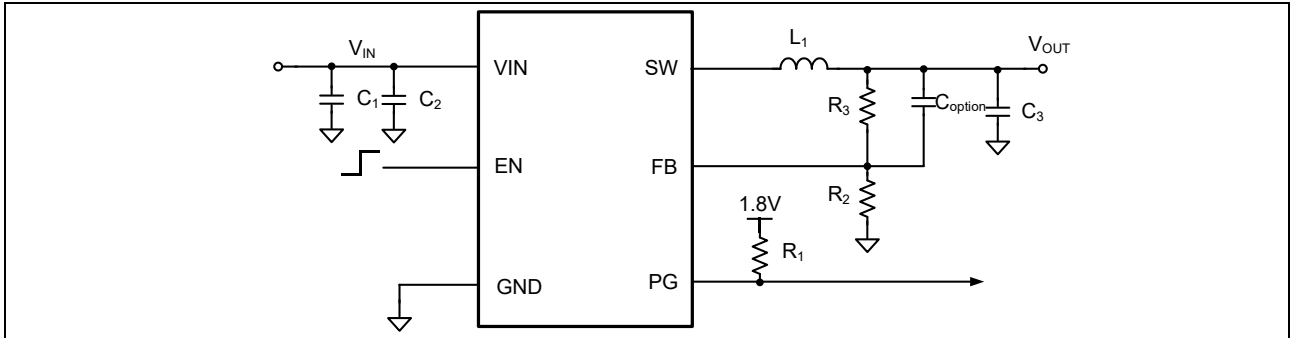


Figure 36. Short Circuit Protection, recovery

Application Circuit



Note*: This application circuit is for reference only.

The ET83102 is a highly efficient, small, and highly flexible synchronous step-down DC-DC converter that is easy to use. A typical system consists of an ET83102H IC, a power inductor L_1 , and several resistors and capacitors as shown in the application circuit above.

Table II. Typical application of 1.2V output voltage

Symbol	Description	Recommended
L_1	1.5 μ H, 2520	DfE252010P-1R5M, Murata (TOKO) HTTH25201T-1R5MSR, Cyntec
C_1, C_3	10 μ F, 6.3V, 0603	GRM188R61A106ME69, Murata
C_2	0.1 μ F, 10V, 0402	GRM022R61A104ME01, Murata
C_{option}	10~100pF, 0402	Reserved
R_1	100k Ω , $\pm 5\%$, 0603	Std.
R_2	100k Ω , $\pm 1\%$, 0603	Std.
R_3	100k Ω , $\pm 1\%$, 0603	Std.

Output Voltage Setting

The output voltage can be programmed by adjusting the external resistor divider R_{UP} and R_{DOWN} according to the equation below:

$$V_{OUT} = \left(\frac{R_{UP}}{R_{DOWN}} + 1 \right) * V_{ref}$$

When the output voltage is regulated, the typical voltage at FB pin is 0.6V

For better accuracy, the R_{DOWN} is recommended to be lower than 100k Ω to ensure the current flowing through R_{DOWN} is at least 100 times larger than the FB pin leakage current. Lower resistor values are recommended for higher accuracy and lower noise at the cost of additional current consumption.

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For a 1.2V-output application, a 100kΩ R_{DOWN} is selected and the R_{UP} is 100kΩ.

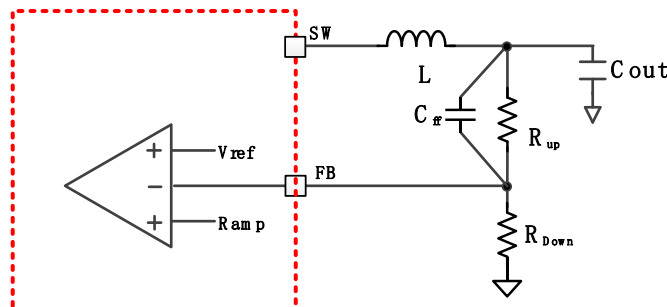


Figure 37. FB connection

Inductor Selection

The inductor is a critical component, related to efficiency, loop stability and transient response, etc. The ET83102 is designed for a nominal 1-μH inductor. Smaller values than nominal inductance cause larger inductor current ripple, which cause larger negative inductor currents in forced PWM mode and higher peak currents at full load.

$$I_{Lp2p} = \frac{(V_{IN} - V_{OUT}) * V_{OUT}}{V_{IN} * L * f_{SW}}$$

Please make sure the minimum peak current is higher than 150mA any time.

Capacitor Selection

The input capacitor not only serves as the power supply of the ET83102H but also filters the high-frequency noise when switching. A 10μF input capacitor is recommended to serve as the bulk input, an additional 0.1~1μF 0201 small size capacitor is strongly recommended to be placed in parallel with the bulk capacitor to filter out the noise.

The output capacitor is another critical component, related to the loop stability and transient response as well. The table lists the typical capacitor and inductor selection and please consider the voltage derating effect when selecting Multi-layer ceramic capacitor (MLCC). At least one 10μF output capacitor should be soldered to guarantee stability. More output capacitors are good for better loop stability and transient performance.

The ET83102 supports 0.7μH and 2.2μH inductors as well, but the light load efficiency is impacted with 0.7μH and transient performance is impacted with 2.2μH inductor. Please verify the efficiency and transient behavior before selecting different inductance components.

Table III. Recommended Inductor and Output Capacitor

Output Voltage	FSW	L	COUT
0.6V	1.5MHz	1μH/2.2μH	≥22μF*2
1.2V			10μF*1
1.8V			10μF*1
3.3V			10μF*1

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PCB Layout Guide

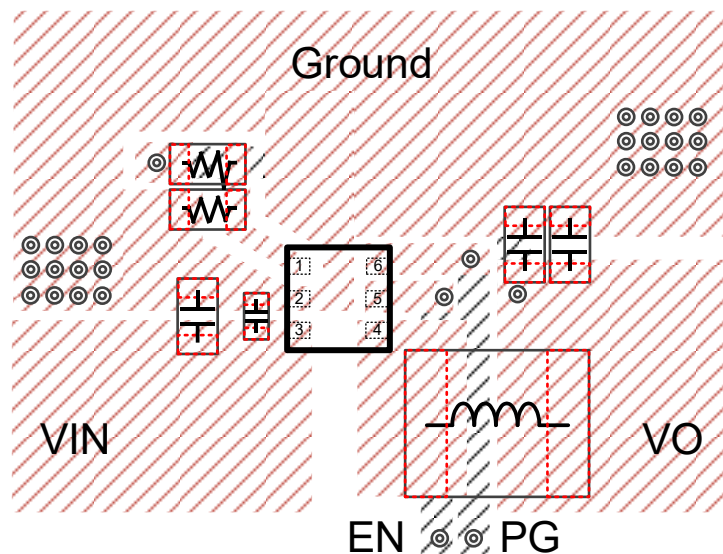


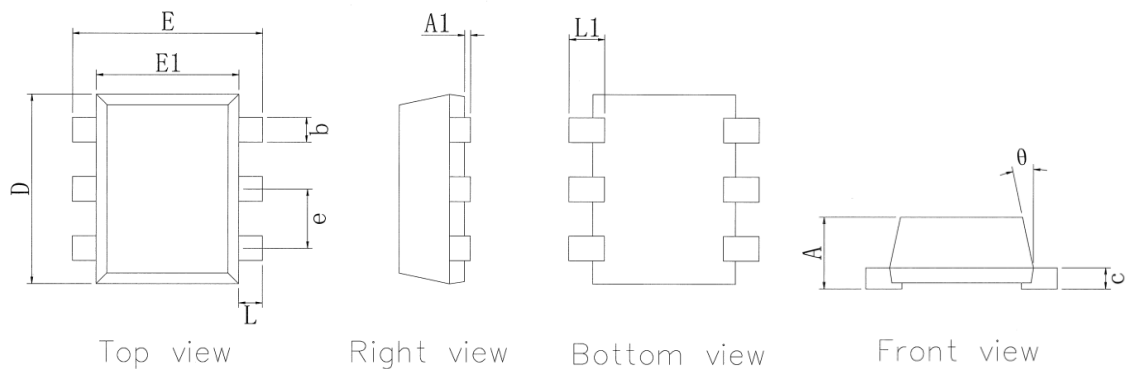
Figure 38 SOT563 Layout example (Top Layer)

Please follow this specific order carefully to achieve the proper layout.

- Place input capacitor C_1 as close as possible to VIN pin and GND pin and use shortest copper trace connection or ground plane.
- Put output capacitor C_2 near to the inductor output terminal and the device. Ground connections need to be tied to the IC ground with a short copper trace or ground plane.
- Place inductor input terminal to SW pin as close as possible and limit SW node copper area to lower electrical and magnetic field radiation. Do not use multiple layers in parallel for this connection. Minimize parasitic capacitance from this area to any other trace or plane.
- At least 2-layer PCB is recommended and with 2-oz thickness copper to achieve good thermal performance.

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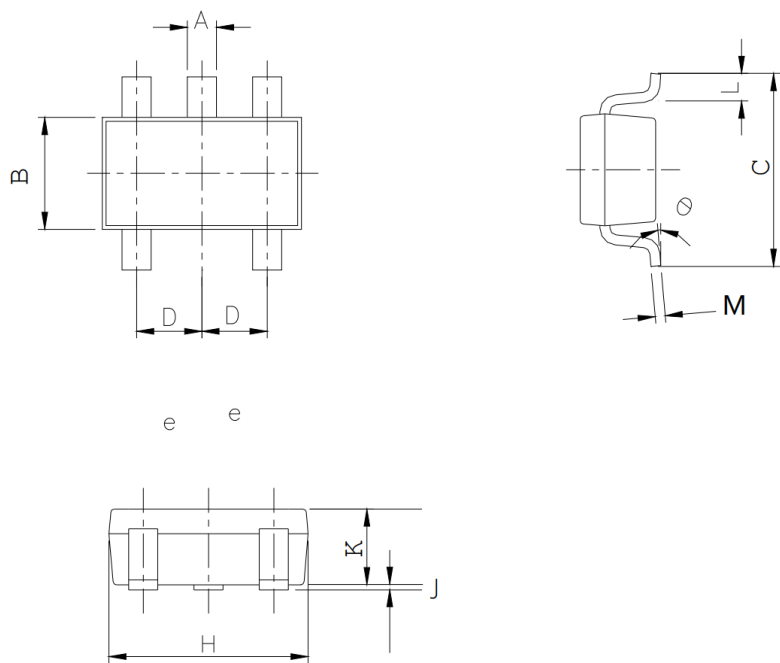
Package Dimension (SOT563)



Dimensions Table (Units: mm)

Symbol	Min	Nom	Max
A	0.530		0.600
A1	0.000		0.050
b	0.170		0.270
c	0.110		0.160
D	1.500	1.600	1.700
E	1.500	1.600	1.700
E1	1.100	1.200	1.300
e	0.500 BSC		
L	0.10	0.20	0.30
L1	0.200	0.300	0.400
theta	11°	12°	13°

Package Dimension (SOT23-5)

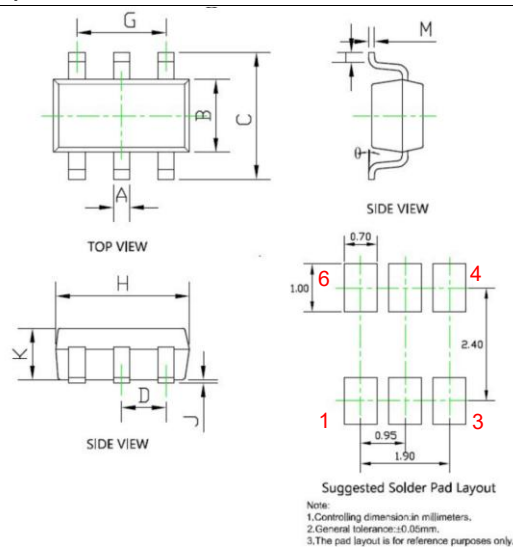


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Dimensions Table (Units: mm)

Symbol	Min	Nom	Max
A	0.300		0.500
B	1.500		1.700
C	2.650		2.950
H	2.820		3.020
J	0.000		0.000
K	1.050		1.150
L	0.300		0.600
D	0.950BSD		
M	0.100		0.200
θ	0°		8°

Package Dimension (SOT23-6)

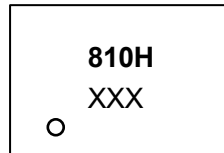


Dimensions Table (Units: mm)

Symbol	Min	Nom	Max
A	0.300		0.500
B	1.500		1.700
C	2.650		2.950
H	2.820		3.020
J	0.000		0.100
K	1.050		1.150
L	0.300		0.600
D	0.950BSC		
G	1.900BSC		
M	0.100		0.200
θ	0°		8°

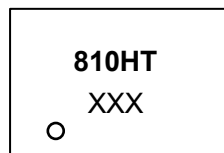
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Marking (SOT563)



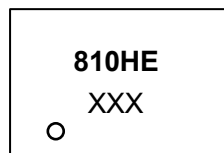
XXX = Tracking Number

Marking (SOT235)



XXX = Tracking Number

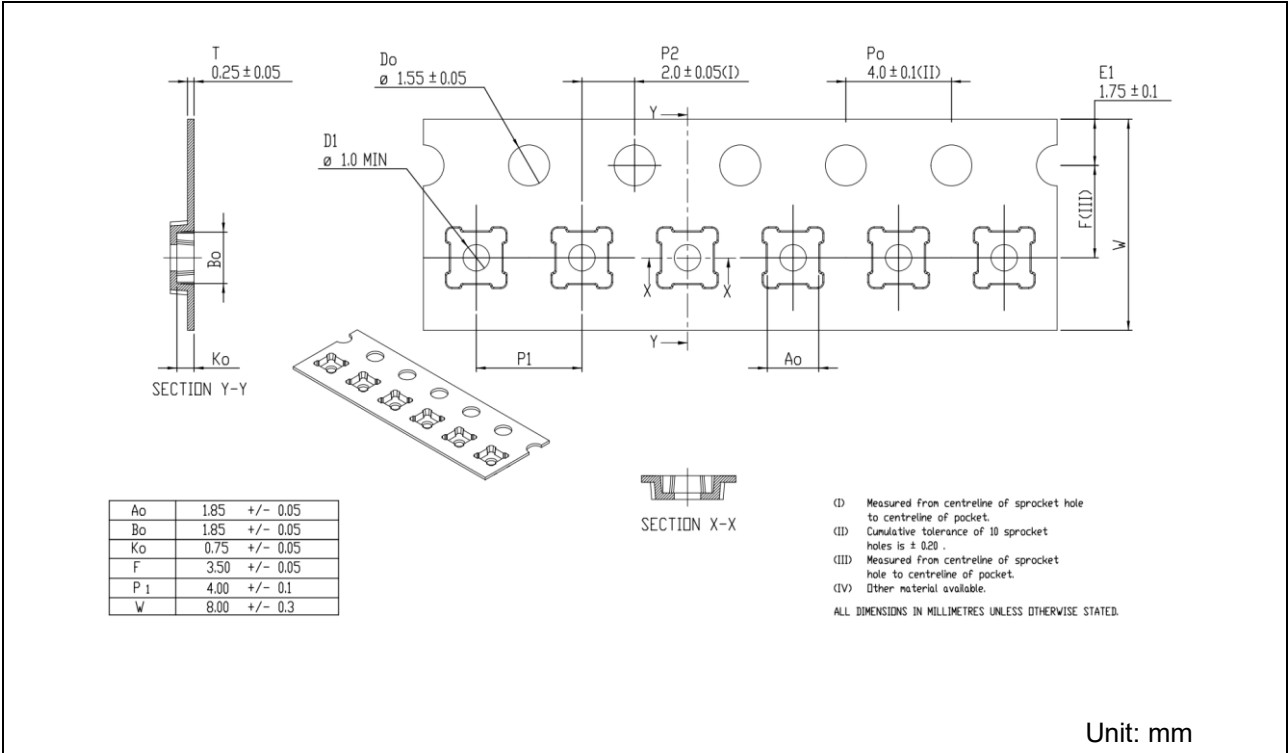
Marking (SOT236)



XXX = Tracking Number

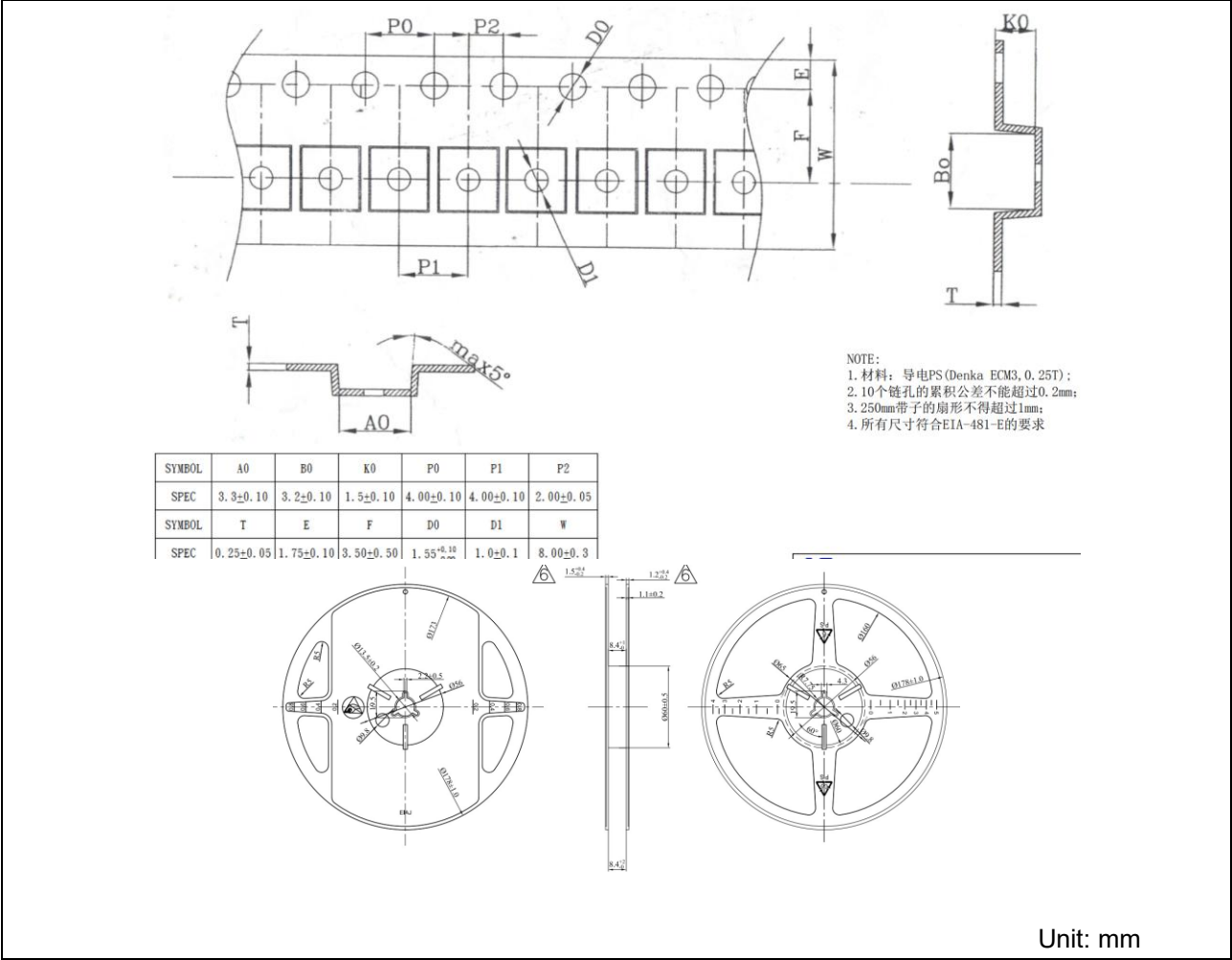
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Tape Information (SOT563)



ET83102H, High Efficiency, 6V/1.5A BUCK Converter

Tape Information (SOT23-5, SOT23-6)



Revision History and Checking Table

Version	Date	Revision Item	Modifier	Function & Spec Checking	Package & Tape Checking
1.0	2026.6.1	First Release	Charles Wong		